UNIVERSITY OF MALAYA MEDICAL CENTRE
NEONATAL INTENSIVE CARE UNIT NEONATAL SEPSIS MANAGEMENT AND ANTIBIOTICS GUIDANCE

Version/ Revisions	Authors	Comments
1.0 Version 2013	Azanna A Kamar, Lai Nai Ming, Choo Yao Mun	First Draft
2.0 Version July 2020	Tan Lay Teng, Azanna A Kamar, Choo Yao Mun, Koh Mia Tuang Department of Paediatrics Sasheela Ponnampanalavanar, Department of Infection Control Jasreena Kaur, Department of Pharmacy	Overview of clinical management. Revisions made based on PPUM NICU antibiogram results
3.0 Version March 2026	Azanna A Kamar, Choo Yao Mun, Chan Kok Joo, Tan Lay Teng, Department of Paediatrics Lim Tse Han, Department of Pharmacy	Change of drug dosages, preferred choice, prophylactic antibiotics.

1.0 OVERVIEW

Infection in neonates is a leading cause of mortality in newborns and a major cause of admission to NICU. The term neonatal sepsis or *sepsis neonatorum* commonly refers to a condition of bacterial, viral or fungal infection associated with haemodynamic changes and other clinical manifestations seen during the neonatal age group or older.

Early onset neonatal sepsis (EONS): ≤72 hours of life.
Late onset neonatal sepsis (LONS) : >72 hours of life.

Note: Definitions of late onset sepsis occurring at more than 7 days have been described in epidemiologic studies and in the description of the clinical course of Group B Streptococcus.

Epidemiology

- The overall incidence of neonatal sepsis varies from 1 to 5 for every 10,000 live births.
- Globally neonatal sepsis accounts between 13 to 26% of total neonatal deaths, with variable estimates of disease burden between various income levels. For example, the incidence of neonatal sepsis in Asia ranges from 7.1 to 38 per 1000 live births in comparison to 6 to 9 cases per 1000 live births in the United States and Australasian countries. A high proportion of newborns in developing countries die at home post-delivery with neonatal sepsis carrying a high impact on the overall newborn mortality.
- Data from the Malaysian neonatal registry reported the incidence of sepsis as 16.1% of very low birth weight (VLBW) infants, where 9.8% was EONS and 90.2% was LONS. High mortality rates of up to 43.8% is observed in infants with GBS EONS. Up to 66.7% of VLBW infants with gram negative sepsis die.

2.0 RISK FACTORS

Risk factors for early onset sepsis:

Maternal GBS (Group B Streptococcus) carrier (high vaginal swab [HVS], rectal swab, urine culture, previous pregnancy of baby with GBS sepsis)

- Prolonged Rupture of Membranes (PROM) (> 18 hours)
- Preterm labour/ Preterm prelabour rupture of membranes (PPROM)
- Maternal pyrexia > 38˚C, maternal peripartum infection, clinical chorioamnionitis, discoloured or foul-smelling liquor, maternal urinary tract infection
- Traumatic delivery
- Birth asphyxia
- Low birth weight
- Confirmed sepsis in co-twin (multiple pregnancy)
- Infant with galactosaemia (increased susceptibility to *E. coli*)

Risk factors for late onset sepsis:

- Any indwelling intravascular access - central venous line, arterial line, umbilical catheter
- Low gestational age
- Low birth weight
- Mechanical ventilation
- Prolonged hospital stay

3.0 AETIOLOGY

Table 1: Aetiology of Neonatal Sepsis

Early-onset Pathogens	**Late-onset Pathogens**
Group B *Streptococcus (Streptococcus agalactiae)* *Escherichia coli* *Klebsiella sp* *Staphylococcus aureus* Coagulase-negative S*taphylococcus* (CONS) *Listeria monocytogenes* Enterococci Other Streptococci: *Streptococcus pyogenes, Streptococcus viridans, Streptococcus pneumoniae* Other gram-negative bacteria: *Haemophilus spp*	Coagulase-negative Staphylococcus (CONS) Group B Streptococcus *(Streptococcus agalactiae)* *Escherichia coli* *Klebsiella* sp. *Staphylococcus aureus* *Candida albicans* Enterococci *Pseudomonas aeroginosa* *Acinetobacter sp* Others : *Enterobacter, Citrobacter, Serratia* Other *Candida* species (Non-albicans Candida) e.g. *C glabrata, C parapsilosus, C kruseo, C tropicalis* *Aspergillus fumigatus, Aspergillus spp*

4.0 ASSESSMENT AND EVALUATION

- Early recognition is essential as sepsis onset is often subtle, with extremely rapid progression. Approximately 1% of well-appearing healthy infants at birth may develop signs of infection occurring after a variable time period.
- Peripartum risk factors place an infant to be at higher risk of sepsis, e.g. prolonged rupture of membranes of more than 18 hours places an infant to be at 10-fold higher risk of sepsis.
- Careful perinatal history, and clinical examination is essential.
- Sepsis risk calculators e.g. Kaiser Permanente study group neonatal sepsis calculator, may be used to calculate probability of sepsis risk.
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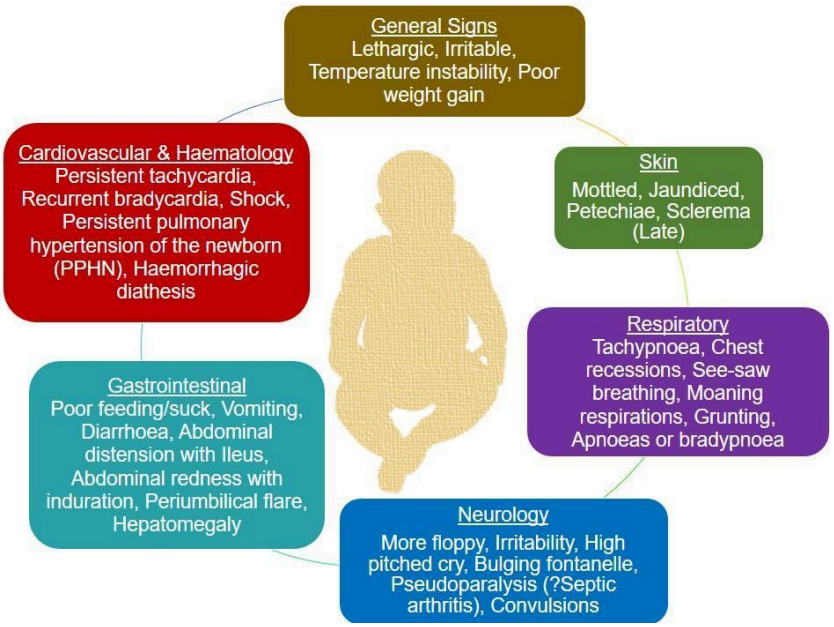


Figure 2: Clinical Features of Neonatal Sepsis

5.0 INVESTIGATIONS

(A) Evidence of Infection and Inflammation

1. **White blood cell count and differentials:** There is poor positive predictive value in total white cell count as an indicator of neonatal sepsis. Low white blood cell count of <5000/mm^3 is associated with the diagnosis of sepsis. Often, the trend of white cell count taken 8 to 12 hours apart is a better indicator. The neutrophils are responsible for digestion of bacteria. However, its chemotaxis may be immature with poor response, especially in preterm infants resulting in release of immature neutrophils. White blood cell parameters which may be used to guide diagnosis include:

    a. Absolute neutrophil count (ANC) – neutropaenia (<1800/mm^3 at birth and <7800/mm^3 at 12 – 14 hours), has better specificity compared with elevated neutrophils. Neutropaenia indicate depletion of the neutrophil storage pool which places the infant to be at risk of dying of sepsis.

b. Immature to total neutrophil ratio (I/T ratio)

$$IT\ Ratio\ =\ \frac{[Metamyelocytes +\ Band\ neutrophils]}{Total\ neutrophils}$$

IT Ratio > 0.20: Infection highly suspicious

IT Ratio > 0.8: High risk of death from sepsis

I/T ratio has the best sensitivity amongst the neutrophil indices with high negative predictive value of up to 99%. However, it has a low positive predictive value of 25%. It indicates the proportion of immature neutrophils (metamyelocytes and band neutrophils) which are released from the neutrophil storage pool which are too immature to effectively ward off infection.

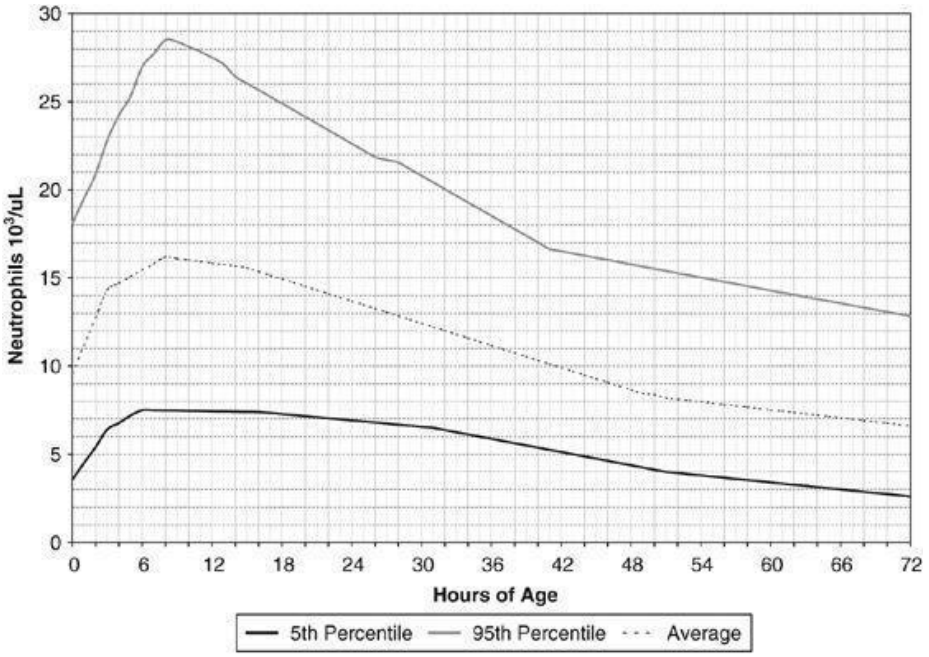


Figure 3: Neutrophil levels of neonates born at ≥ 36 weeks' gestation during the first 72 hours at birth.

(Source : Schmutz, N., Henry, E., Jopling, J., & Christensen, R.D. (2008). Expected ranges for blood neutrophil concentrations of neonates: the Manroe and Mouzinho charts revisited. Journal of Perinatology, 28, 275-281.)

2. **Platelet Count:** Low platelet count is a common manifestation of neonatal sepsis. This is more commonly associated with gram positive organism sepsis. The phenomenon may still be observed with gram negative organisms. The mechanism is uncertain,but is said to be caused by bacterial endothelial damage leading to platelet adhesion and aggregation. It has also been shown to be caused by the presence of circulating immune complexes in the septic neonate.

3. **Blood culture and sensitivity:** A positive culture growing from blood or sterile body fluid remains the criteria in diagnosing neonatal sepsis. Up to two-thirds of younger infants <2

months of age have colony forming units of <10 CFU>ml. Therefore, an optimal amount of blood of 1 ml should be drawn for the purpose of culture or else cultures may remain negative. Cultures may also remain negative in view of prenatal antibiotics administration. It may be difficult to determine whether positive culture is a contaminant, especially if coagulase negative Staphylococcus (CONS) is grown. Term infants without any indwelling catheters are less likely to grow CONS.

4. **Lumbar puncture:** In bacteraemic infants, the incidence of associated meningitis is as high as 23%. Therefore, performing a lumbar puncture for cerebrospinal fluid (CSF) examination (gram stain, culture, cell count and biochemistry – protein and glucose concentration), may be necessary in an infant with a positive blood culture, abnormal laboratory data, or in infants with a clinical course that is strongly suggestive of sepsis. If there is already a strong suspicion of clinical infection or meningitis from the outset, a CSF sample is best obtained before commencement of antibiotics but should not result in undue delay in antibiotics initiation.

Table 2: Normal CSF Ranges in Neonates

	RBC/mm^3	**Median WBC/µL (IQR)**	**Median Protein g/L (IQR)**
Term ≤ 7 days	9 (0-50)	3 (1-6)	0.78 (0.60-1.0)
Term >7 days	<10	2 (1-4)	0.57 (0.42-0.77)
Preterm ≤ 7 days	30 (0-333)	3 (1-7)	1.16 (0.93-1.38)
Preterm > 7 days	30	3 (1-4)	0.93 (0.69-1.22)
ELBW ≤ 7 days		3 (1-8)	1.62 (1.15-2.22)
ELBW > 7 days		4 (0-14)	1.59 (0.95 – 3.70)
"Bloody Tap": Calculate WBC to RBC ratio. Allow 1:500, if unsure, to treat as meningitis. CSF glucose: Blood glucose ratio ≥ 0.6; CSF glucose ≥ 2.5 mmol/L. *ELBW: Extremely low birth weight. Preterm= Infants <37 weeks gestation.*			

Adapted from: Chess PR 2019, Avery's Neonatology Board Review. Srinivasan et al., J Pediatr 2012. Rodriguez et.al., J Pediatr 1990. Royal Berkshire NHS Sepsis Guidelines 2018.

5. **C-Reactive Protein (CRP):** CRP is an acute phase reactant produced in the liver. Approximately 10 to 12 hours is required for CRP to change significantly after the onset of an

infection with low sensitivity during early stages, and is therefore a late indicator of sepsis. A sequential analysis is more useful in interpretation compared with a single reading. Bacterial sepsis is less likely in infants with persistently low CRP of <1 mg/dL and is therefore often used as a guide to discontinue antibiotics therapy. Caution in interpretation should be taken especially in preterm infants where the response may be less apparent or in infants with meconium aspiration, intraventricular haemorrhage or perinatal asphyxia where the CRP may be responsively raised.

6. **Procalcitonin (PCT):** Release of PCT by parenchymal cells is triggered by bacterial toxins and are therefore a useful marker in neonates. Although the response is more rapid than CRP where its concentration increases within 2-4 hours with peaks at 6-8 hours post-exposure, like CRP, a sequential or serial reading is more useful. In very low birth weight infants, PCT value of >2.4 ng/mL indicates a high risk of neonatal sepsis. Like CRP, it can also be falsely elevated in non-infective conditions. If used in conjunction with CRP, a positive level increases the diagnostic value up to 92%.

(B) Evidence of Multiorgan System Disease

The following may be performed based on clinical suspicion of associated organ involvement.

1. Blood gas analysis and lactate: Should be performed to determine extent of condition. Metabolic acidosis with lactic acidaemia indicating progression to anaerobic metabolism.

2. Renal function, liver function tests: Impairment indicating possible involvement.

3. Haematologic indices and coagulation profile: Manifested as anaemia and thrombocytopaenia, as well as disseminated intravascular coagulopathy.

4. Chest X-ray and/or Abdominal X-ray: If accompanying respiratory or abdominal signs are seen.

5. Liver and renal ultrasound – in cases of persistent sepsis

6. Echocardiography - In cases of persistent sepsis

6.0 MANAGEMENT

Preventive strategies have been described earlier with emphasis on recognition of peripartum risk factors. Initiation of antibiotics must be prompt. Recognition of complications before progression to organ failure may be difficult, but important as progression may be rapid.

Aims of Management

1. Early recognition with timely initiation of treatment
2. Supportive therapy with careful monitoring of complications
    a. Prevent development of septic shock
    b. Recognise and halt progression of complications

(1) Early Recognition with Timely Initiation of Treatment

Initiation of antibiotics

- Consideration of early or late-onset presentation as well as exposures e.g. nosocomial infection determines the choice of antibiotics. See antibiotic guidance in section 7.0

- Avoid undue delay in antibiotics initiation and initial empiric therapy should be initiated with penicillin and aminoglycoside for EONS. However, in babies determined not to have sepsis, antibiotic exposure should be minimised.

- Third generation cephalosporin should only be reserved for suspected gram- negative meningitis. The emergence of cephalosporin resistant organisms (e.g. Enterobacter cloacae, Klebsiella, and Serratia sp) can occur when cefotaxime is routinely used.

- Blood culture results should be traced within 48 to 72 hours and antibiotics regimen adjusted based on susceptibility pattern. Therapy may be discontinued should the results return as negative with no other parameters to support the diagnosis of sepsis.

Adjunctive Treatment Strategies

• Exchange transfusion may be considered should sepsis be accompanied with late features such as sclerema neonatorum. A few studies have demonstrated that exchange transfusion is associated with fewer mortality in the exchange group compared to those without exchange transfusion amongst infants who have developed sclerema neonatorum. However, its use must be cautioned against possible complications. There are few studies regarding this procedure for sepsis.

• IVIg augments antibody dependent cytotoxicity with improvement on neutrophilic function. However, sufficient evidence for its use in reducing death in neonatal sepsis remains unproven.

(2) Supportive Therapy

- Sepsis with rapid progression of its associated complications is termed fulminant sepsis, which may lead to death within 48 hours of its onset.

- Infants who acquire nosocomial sepsis, of lower gestation (extremely low gestational age/ELGAN), and lower birth weight (extremely low birth weight/ELBW <1000grams) are at higher risk of fulminant sepsis.

- Complications associated with fulminant sepsis include:

    - Worsening respiratory status
    - Haemodynamic instability leading to septic shock
    - Cardiac dysfunction
    - Coagulopathy
    - Multiple organ dysfunction

  General supportive therapy aims to prevent progression of fulminant sepsis by ensuring adequate ventilation and maintenance of cardiovascular function.

Strategies to prevent multiorgan dysfunction include:

- *Maintenance of the thermoneutral environment:* Maintenance of the thermoneutral environment is required to prevent the sequelae of anaerobic metabolism which worsens lactate acidaemia. The septic neonate often develops temperature instability due to the infant's inability to auto-regulate the body temperature. Therefore, close temperature monitoring and provision of adequate warmth which may be provided via servo/auto-control warmer should be initiated.

- *Provision of adequate cardiovascular and haemodynamic support:* An infant who has been determined to have sepsis should have close monitoring of vital signs and blood pressure. Timely intervention with crystalloids is required with signs of intravascular volume depletion, such as poor perfusion, prolonged capillary refill time, poor pulse volume and increasing tachycardia. There is insufficient evidence to support aggressive and repeated volume expansion in the hypotensive preterm neonate due to its possible association with intraventricular haemorrhage. Therefore, after initiation of fluid bolus, the neonate who remains hypotensive may be supported with inotropic support. Use of haemodynamic monitoring systems such as the use of a specialised heart rate observation monitoring equipment or specialised systems may be considered.

- *Provision of adequate respiratory support:* Septic infants often will have accompanying pneumonia and/or recurrent apnoea with poor respiratory effort. Ensure timely and adequate provision of respiratory supportive therapy and oxygenation e.g. intubate the infant with repeated apnoeic episodes.

- *Glucose monitoring:* Glucose is an essential brain fuel and the increasing metabolic demands can result in hypoglycaemia in the septic neonate. Conversely, the resultant increase in production of stress hormones such as adrenaline, cortisol and glucagons result in hyperglycaemia.

- *Monitoring of urine output:* Strict charting of the infant's input and output is required to monitor for evidence of renal impairment. Fluids should be adjusted accordingly based on the infant's haemodynamic and renal status.

- *Organ Dysfunction Monitoring:* Clinical monitoring of the progress and severity of organ dysfunction may be considered in a neonate with sepsis. Besides the above and use of specific biomarkers, sepsis scoring systems may be used, enabling prognostication, benchmarking of outcomes, and targeted therapy. These sepsis scores include the following:

    - Neonatal Sequential Organ Failure Assessment (nSOFA) Score
    - Pediatric Logistic Organ Dysfunction (PELOD) Score
    - Neonatal Multiple Organ Dysfunction Score (nMODS), and other similar scores.

  Artificial intelligence tools developed may be used to monitor progress. However, several of these tools are presently still being trialled without sufficient validation for the neonatal population.

7.0 ANTIBIOTICS GUIDANCE

The following tables cover the major indications, duration and dosages of common antibiotics used in the neonatal intensive care unit. The list is not exhaustive, and management strategies for other specific infections need to be discussed with the specialist in-charge.

The Four Moments/Principles of Antibiotics Decision and Administration

1. Identify source of infection
2. Ensure blood cultures and cultures from appropriate anatomical site performed and commence empiric therapy
3. Reassess at 48-72 hr:
    a) Stop: unless positive cultures are reported, or the baby remains unwell.
    b) Targeted therapy: based on culture results
    c) Narrow spectrum: based on clinical response (based on site of infection and if culture results negative (or no culture taken)
4. Duration of antibiotics based on the site of infection and types of microorganism isolated

Escalation of therapy: consider escalation of therapy if persistent or worsening signs of infection or deranged sepsis parameters after 48 -72 hours of adequate antibiotic therapy and source control.

All preterm infants in NICU receiving Cefepime, Cefotaxime, Piperacillin-Tazobactam, Meropenem or broad-spectrum antibiotics, may be considered for commencement of oral prophylaxis nystatin for the length of the antibiotic course, regardless of enteral feeding status.

7.1 Major Indications for the Use of Antibiotics in the NICU

Indications	Preferred	Alternative	Comments
Early onset sepsis Commonest organisms: *Group B Streptococcus* Gram negative rods (*E. coli, Klebsiella sp.*) Less common: *Staphylococci* *Streptococci Enterococci* *Haemophilus influenza* *Listeria monocytogenes*	C-Penicillin AND Gentamicin*	C-Penicillin AND Cefotaxime	Targeting infecting organism acquired antenatally or intrapartum (The commonest organisms are from the vaginal flora). Ampicillin is preferred choice and to replace Penicillin if *Listeria* infections suspected. Cefotaxime should be used judiciously in the neonatal unit. It is associated with increased rates of colonization with strains resistant to the empirical therapy, with higher rates of Extended Spectrum Beta Lactamase (ESBL) infections, invasive candidiasis and death. However, it should be considered in infants with meningitis.

Late onset sepsis Coagulase negative *Staphylococci, Staphylococcus aureus* Gram negative rods (*Pseudomonas aeruginosa*) *Enterococci* *Candida spp.*	C-Penicillin OR Cloxacillin AND Gentamicin* OR Amikacin*	C-Penicillin OR Cloxacillin AND Cefotaxime	Cloxacillin may be indicated if there are risk factors for sepsis from skin organisms, e.g. multiple venepuncture or other invasive procedures. Reserve cephalosporin ONLY for treatment of meningitis. Consider adding cefotaxime if meningitis is suspected. Cefotaxime should be used judiciously in the neonatal unit. It is associated with increased rates of colonization with strains resistant to the empirical therapy, with higher rates of Extended Spectrum Beta Lactamase (ESBL) infections, invasive candidiasis and death. Ceftriaxone should be avoided in neonate because it displaces bilirubin from albumin binding sites, resulting in a higher free bilirubin with subsequent blood brain barrier penetration.
Hospital acquired/ nosocomial sepsis	Vancomycin* AND Cefepime	**Escalation of therapy**: Vancomycin AND Piperacillin-Tazobactam OR Meropenem	Targeting nosocomial infections and translocation of organisms across the immature gut wall. Associated with indwelling lines, catheters and ET tubes. Piperacillin-Tazobactam is a reasonable second line option in pneumonia & intraabdominal sepsis (non-CONS sepsis with good coverage against Gram positive, Gram negative & anaerobes) with good sensitivity for *Klebsiella, E.coli* and moderate sensitivity for *Acinetobacter* from UMMC antibiogram. Cefepime is the preferred agent when there are Gram negative bacterial with extended spectrum cephalosporin resistance due to AmpC-β-lactamases (Antibiogram shows reduced sensitivity for Cefotaxime to *Acinetobacter baumannii* and *Escherichia coli*). Targeted therapy for *Acinetobacter baumannii* is Ampicillin-Sulbactam.
Meningitis	C-Penicillin AND Cefotaxime	Hospital Acquired: Vancomycin* AND Meropenem	Consider Vancomycin and Meropenem in hospital acquired sepsis with meningitis.

Ventilator-associated/ Nosocomial pneumonia	Target the recent tracheal secretion or throat swab culture result, if negative, target the most prevalent organisms in the unit. If there is no specific pattern of colonization or infection, follow the regime for hospital acquired infection.		
Catheter related blood stream and local skin and soft tissue infections	Vancomycin* AND Cefepime		Vancomycin is preferred for catheter related blood stream infection, but to descalate therapy if culture is Methicillin Sensitive *Staphylococcus aureus* (MSSA).
Necrotising enterocolitis and other intra-abdominal sepsis *Klebsiella spp.* *Escherichia coli Clostridia spp. Coagulase negative Staphylococci Enterococci Bacteroides*	First 7 days of life without previous antibiotics exposure: Ampicillin AND Gentamicin* AND Metronidazole > 7 days of life: Piperacillin-Tazobactam OR Cefepime WITH Vancomycin if suspected MR CONS* AND Add Metronidazole if Stage 3 and above	**Escalation of therapy** *Meropenem or WITH or WITHOUT Vancomycin*	Use Vancomycin instead of ampicillin for suspected MRSA and ampicillin-resistant enterococcal infection. Piperacillin-Tazobactam can be used for non-CNS systemic infections, necrotising enterocolitis and intra-abdominal infections caused by susceptible Gram positive and Gram negative bacteria including anaerobes and many Enterobacterales and Pseudomonas spp. Concurrent use with vancomycin has been suggested to be associated with an increased incidence of acute kidney injury in adults and children but unclear in neonates. Meropenem has the propensity to select out multi-drug response organisms. To be used if persistent signs of infection or deranged sepsis parameters after 48 -72 hours of adequate antibiotic therapy and source control. Consider adding antifungal if the patient continues to deteriorate despite being on appropriate antibiotics or second line therapy.
Fungal sepsis	Fluconazole	**Escalation of therapy** Lipid complex amphotericin B	In infants with persistent signs, deranged sepsis parameters, and without improvement in clinical symptoms despite adequate antibiotic coverage and source control, fungal sepsis should be considered. Risk factors for fungal sepsis including use of TPN, gut perforation/ abdominal surgical intervention, mechanical ventilation and prolonged use of broad spectrum antibiotics.

7.2 General Guidance on the Recommended Duration of Antibiotic Treatment

Types of Infection	Days
Suspected sepsis	If negative blood culture, initial clinical suspicion not strong, reassuring clinical condition & low CRP, consider stopping antibiotics at 48 hours If negative blood culture, very strong clinical suspicion of sepsis or elevated CRP, consider 5 days of antibiotics
Gram Positive Bacteraemia	Repeat blood culture at 48 to 72 hours of antibiotic. At least 7 days of antibiotics as guided by microorganism, clinical response, CRP and repeated blood culture. Coagulase negative staphylococcal bacteraemia: 7 days if line-related and line retained Staphylococcus aureus bacteraemia (uncomplicated – absence of osteomyelitis, endocarditis or deep seated source of infection): 14 days (longer course required if complicated) Group B streptococcal bacteraemia without meningitis: 10 days Listeria bacteraemia without meningitis: 14 days
Gram Negative Bacteraemia	Repeat blood culture at 48 to 72 hours of antibiotic. At least 7-10 days of antibiotics as guided by microorganism, clinical response, CRP and repeated blood culture.
Fungaemia	Repeat blood culture every 72 hours of antifungal therapy. Screening for invasive fungal infection including opthalmological examination and ultrasound abdomen. At least 2 weeks of antifungal from the time of negative blood culture.
Meningitis Gram positive meningitis Gram negative meningitis Listeria meningitis Culture negative/partially-treated	Repeat CSF biochemistry and culture at 48 hours of antibiotics. 14 days 21 days 21 days 14 days
Pneumonia Congenital and community acquired Hospital-acquired	 5 – 10 days depending on screening and extent of infection. 7 – 14 days depending on screening and extent of infection.

Necrotising enterocolitis Stage IA and IB Stage IIA Stage IIB Stage IIIA Stage IIIB	Depending on clinical course 3 days 7 - 10 days 14 days 14 days At least 14 days
Catheter related blood stream infections	As in bacteraemia. At least 7 to 10 days of antibiotics as guided by microorganism, clinical response, CRP and repeated blood culture. Remove catheter (source control) if persistent bacteraemia.
Joint infection (Septic arthritis/ osteomyelitis)	4 – 6 weeks
Other specific infections such as urinary tract, skin and eye infections	5 – 10 days

7.3 Recommended Dosages for Common Antibiotics Prescribed in Neonates

Drug Name	Postmenstrual Age	Postnatal Day	Dose
Benzylpenicillin	< 30 weeks	0 – 28 days	100,000 units/kg/dose IV q12h
		> 28 days	100,000 units/kg/dose IV q8h
	30 – 36 weeks	0 – 14 days	100,000 units/kg/dose IV q12h
		> 14 days	100,000 units/kg/dose IV q8h
	37 – 44 weeks	0 – 7 days	100,000 units/kg/dose IV q12h
		> 7 days	100,000 units/kg/dose IV q8h
	≥ 45 weeks	All	100,000 units/kg/dose IV q6h

Ampicillin	< 30 weeks	0 – 28 days	50mg/kg/dose IV q12h
		> 28 days	50mg/kg/dose IV q8h
	30 – 36 weeks	0 – 14 days	50mg/kg/dose IV q12h
		> 14 days	50mg/kg/dose IV q8h
	37 – 44 weeks	0 to 7 days	50mg/kg/dose IV q12h
		> 7 days	50mg/kg/dose IV q8h
	≥ 45 weeks	All	50mg/kg/dose IV q6h
Gentamicin*	< 30 weeks		5 mg/kg/dose IV q48h
	30 – 34 weeks		4 mg/kg/dose IV q36h
	≥ 35 weeks		4 mg/kg/dose IV q24h
Amikacin*	< 30 weeks		7.5 mg/kg/dose IV q24h
	30 – 34 weeks		10 mg/kg/dose IV q24h
	≥ 35 weeks		15 mg/kg/dose IV q24h
Cefotaxime	< 30 weeks	0 – 28 days	50mg/kg/dose IV q12h
		> 28 days	50mg/kg/dose IV q8h
	30 – 36 weeks	0 – 14 days	50mg/kg/dose IV q12h
		> 14 days	50mg/kg/dose IV q8h
	37 – 44 weeks	0 to 7 days	50mg/kg/dose IV q12h
		> 7 days	50mg/kg/dose IV q8h
	≥ 45 weeks	All	50mg/kg/dose IV q6h

Metronidazole	Loading dose Maintenance dose < 27 weeks 27 – 33 weeks 34-40 weeks ≥ 41 weeks	 0 – 26 days 0 – 14 days 0 to 7 days All	15mg/kg/dose IV x 1 7.5mg/kg/dose IV q24h 7.5mg/kg/dose IV q12h 7.5mg/kg/dose IV 8 hourly 7.5mg/kg/dose IV 6 hourly
Piperacillin + Tazobactam	< 30 weeks 30 – 36 weeks 37 – 44 weeks ≥ 45 weeks	0 – 28 days > 28 days 0 – 14 days > 14 days 0 to 7 days > 7 days All	100mg/kg/dose IV q12h 100mg/kg/dose IV q8h 100mg/kg/dose IV q12h 100mg/kg/dose IV q8h 100mg/kg/dose IV q12h 100mg/kg/dose IV q8h 100mg/kg/dose IV q8h
Cefepime	All	All	40 mg/kg/dose q8h
Meropenem	< 32 weeks ≥ 32 weeks	< 14 days ≥ 14 days < 14 days ≥ 14 days	20-40 mg/kg/dose q12h 20-40 mg/kg/dose q8h 20-40 mg/kg/dose q8h 30-40 mg/kg/dose q8h

Cloxacillin	All	0 to 7 days 8 to 20 days ≥ 21 days	25-50mg/kg/dose IV q12h 25-50mg/kg/dose IV q8h 25-50mg/kg/dose IV q6h *Use 50mg/kg/dose for moderate to severe infection
Ampicillin + Sulbactam *Preferred to use as targeted therapy for Acinebacter baumanni. Considering higher dose for Extensively Drug-Resistant(XDR) organisms	< 37 weeks ≥ 37 weeks Infants		IV 100mg/kg/day (ampicillin component) divided every 12 hours IV 100mg/kg/day (ampicillin component) divided every 8 hours IV 100-200mg/kg/day (ampicillin component) divided every 6 hours

Vancomycin
Loading dose: Birth weight < 1 kg - 15 mg
Birth weight > 1 kg - 15 mg/kg

Post Menstrual Age (weeks)	Appropriate For Gestational Age	Small For Gestational Age < 10th Centile
< 26	12.5 mg/kg daily	10 mg/kg daily
26 to < 27	15 mg/kg daily	12.5 mg/kg daily
27 to < 28	15 mg/kg daily	12.5 mg/kg daily
28 to < 29	10 mg/kg 12 hourly	15 mg/kg daily
29 to < 30	10 mg/kg 12 hourly	15 mg/kg daily

30 to < 31	12.5 mg/kg 12 hourly	10 mg/kg 12 hourly
31 to < 32	12.5 mg/kg 12 hourly	10 mg/kg 12 hourly
32 to < 33	15 mg/kg 12 hourly	10 mg/kg 12 hourly
33 to < 34	15 mg/kg 12 hourly	12.5 mg/kg 12 hourly
34 to <37	Loading dose 20 mg/kg then 20 mg/kg 12 hourly	15 mg/kg 12 hourly

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*** Blood for therapeutic drug monitoring must be performed if gentamycin, amikacin and vancomycin are chosen as the therapeutic agent of choice. Closer monitoring may especially be required in infants with hypoxia/ encephalopathy and those undergoing therapeutic hypothermia. The dosage may be changed in accordance to the drug's peak and trough blood levels.**

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